MIT SCHOOL OF COMPUTING
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MIT-ADT UNIVERSITY
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A Leap Towards World Class Education

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Subject: P.L Roll No.: 32 Exp. No.:
Name of the Experiment: Assignment - 2

Marks	Teacher's Signature with date
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Performed on:
Submitted on:

Assignment - 2

Q1) List the python sequence data types and state one key characteristics of each with respect to mutability or ordering.

Ans) → Data types :-

a) String (str) :→ immutable and ordered collection of characters

b) List :→ mutable and ordered collection of elements

c) Tuple :→ immutable and ordered collection of elements

d) Range :→ immutable and ordered sequence of numbers

Q2) Explain how indexing and slicing work in Python sequences. Illustrate how they differ when applied to strings & lists.

Ans) Indexing is used to access a single element of a sequence using its position. Positive indexing starts from -1.

Slicing extracts a part of sequence using the syntax :-

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

Difference in strings and lists :-

↳ In strings, indexing / slicing returns characters or sub-strings.

↳ In lists, indexing / slicing returns elements or ~~sublists~~ sub-lists.

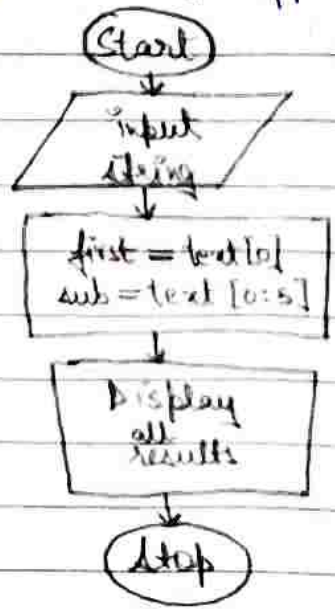
Eg :- string :- "Python" [1:4] → "yth"
list :- [1, 2, 3, 4] [1:3] → [2, 3]

Q3) Python program using indexing, slicing & built-in functions.

Ans) → Code :-

```
text = "Engineering"
first_ch = text[0]
sub_text = text [0:5]
length = len(text)
upper_text = text.upper()

print ("First character :", first_ch)
print ("Sliced text :", sub_text)
print ("length :", length)
print ("uppercase :", upper text)
```



Q4) Difference between equality (==) and identity (is).

Ans) i) == (equality operator) :→ checks whether two sequences have the same values

ii) is (identity operator) :→ checks whether two variables refer to the same object in memory

Eg:- a = [1, 2]
b = [1, 2] → a == b is True but a is b is False

Q5) Evaluation of lists, sets and dictionaries for students records.

Ans) i) Lists allow duplicates and are ordered, but searching is slower

ii) Sets store only unique values and are fast for membership testing.

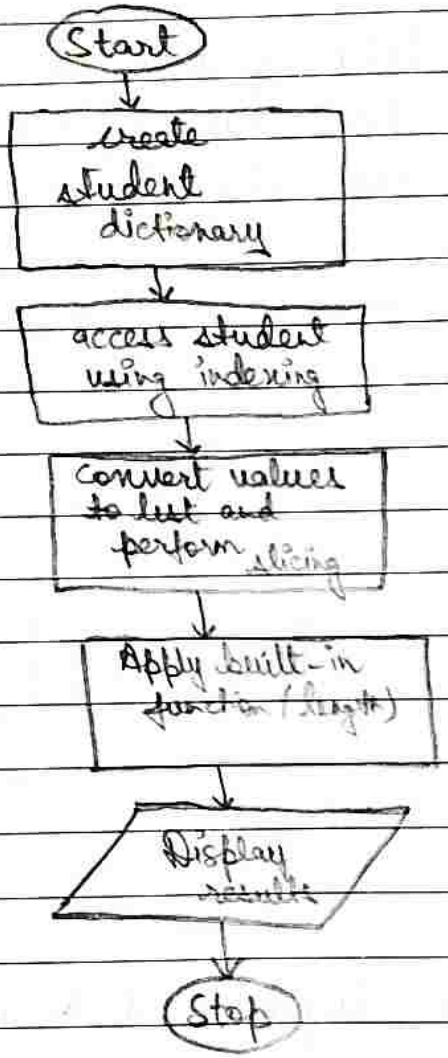
iii) Dictionaries store data in key-value pairs and allow fast access using keys

- Most efficient :- dictionaries are best for student records because each student can be stored with a unique ID as a key, enabling fast lookup, update and retrieval

Q6) Program to manage student data

Ans)
```
students = {
    101: "Aman"
    102: "Riya"
    103: "Aran"
}
```

```
print ("Student with ID 101 :", students [101])
names = list( student. values ())
print ("First two students :", names [0,2])
print ("Total students :", len (students))
```



Q7) Print first and last character of a string.

Ans) • Algorithm :-

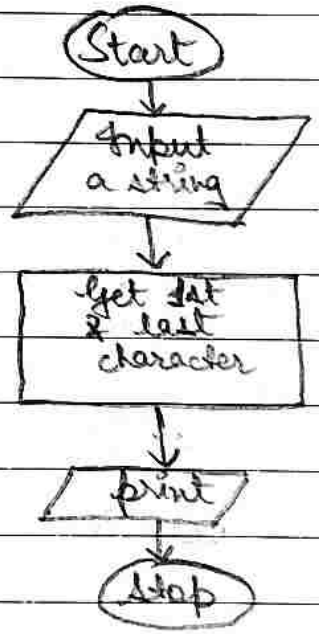
i) Start
ii) Input a string
iii) Print 1st character using index 0
iv) Print last character using index -1
v) Stop

- Code :-

```
text = input ("Enter a string :")
print ("First character :", text [0])
print ("Last character :", text [-1])
```

- O/P :- Enter a string = Python
  First character = P
  Last character = n

- Flowchart :-



Q8) Create list, tuple, set & dictionary

Ans) • Algorithm :-

i) Start
ii) Create list, tuple, set, dictionary
iii) Display them
iv) Stop

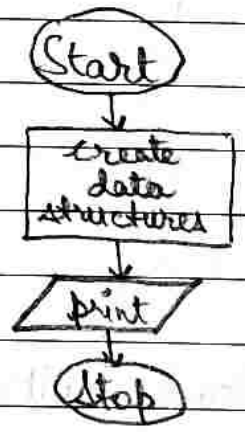
- Code :-

```
list = [1, 2, 3]
tup = (4, 5, 6)
st = {7, 8, 9}
dct = {"a":1, "b":2}
```

- O/P :-

```
list = [1,2,3]
Tuple = (4, 5, 6)
set = {7, 8, 9}
dictionary = {"a":1, "b":2}
```

- Flowchart :-



Q9) Program to understand indexing.

Ans) • Algorithm :-

i) Start
ii) Create a list
iii) Access elements using indexing
iv) Print elements
v) Stop

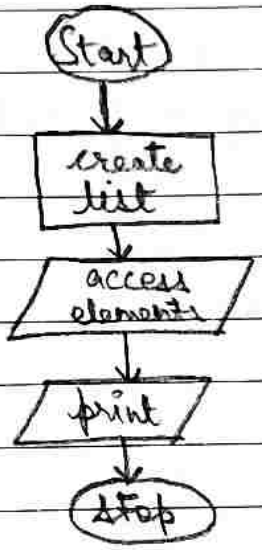
- Code :-

```
numbers = [10, 20, 30, 40]
```

```
print("first element:", numbers[0])
print("second element:", numbers[1])
print("last element:", numbers[-1])
```

- O/P :- first element = 10
  second element = 20
  last element = 40

- Flowchart :-



Q10) Program to understand built-in functions

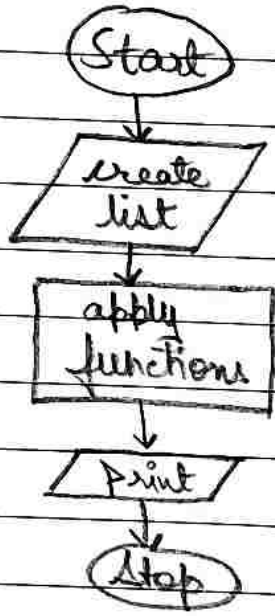
Ans) • Algorithm :-

i) Start
ii) Create a list
iii) Apply built in functions
iv) Print results
v) Stop

- Code :-

```
numbers = [5, 10, 15, 20]
print("length:", len(numbers))
print("Maximum:", max(numbers))
```

```
print ("Minimum :", min (numbers))
print ("Sum :", sum (numbers))
```

- O/P :- Length = 4
  Max = 20
  Min = 5
  Sum = 50

- Flowchart :-